

Geethanjali College of Engineering and Technology

(UGC AUTONOMOUS INSTITUTION)

(Accredited by NBA and NAAC with 'A' grade, Approved by AICTE New Delhi and Affiliated to JNTUH)

Cheeryal (V), Keesara (M), Medchal (Dist), Telangana – 501 301.

DEEP LEARNING LAB (20CS41L08)

Laboratory Workbook

IV Year I Semester (AR22)



Geethanjali

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

2026-2027

Lab – In charge

HOD CSE-AIML

Dr. A. Nageswar Rao

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DEEP LEARNING LAB

(20CS41L08)

Laboratory Workbook

Name : _____

Roll No : _____ Class: _____ Branch: _____

Academic Year: 20____ - 20_____

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CERTIFICATE

*This is to certify that Mr. / Ms. _____ has
satisfactorily completed _____ number of
experiments in the Deep Learning Laboratory.*

Roll No: _____

Branch: _____ Section: _____

Year: _____

Academic Year: _____

Head of the Dept.

CSE-AIML

Faculty Incharge

Internal Examiner

External Examiner

LIST OF LAB EXERCISES

Deep Learning Lab

S.No.	Name of the Program	CO'S	PO'S	P. No.
1	Demonstrate normalization of input data, basic activation functions such as the softmax, sigmoid, dsigmoid, etc	CO1	PO1,PO2,PO3,PO4,PO9	
2	Build a neural network for logistic regression to minimize the cost function and update the parameters	CO1	PO1,PO2,PO3,PO4,PO9	
3	Implement backward propagation neural network for a two- class classification with a single hidden layer, non-linear activation function like tanh and compute the cross-entropy loss.	CO2	PO1,PO2,PO3,PO4,PO9	
4	Build a deep neural network with more than one hidden layer, non-linear functions like ReLU.	CO2	PO1,PO2,PO3,PO4,PO5,PO9	
5	Build deep neural network to any classification problem and compare its accuracy to logistic regression	CO2	PO1,PO2,PO3,PO4,PO5,PO9	
6	Apply Regularization techniques in deep learning model with backward propagation.	CO3	PO1,PO2,PO3,PO5,PO9	
7	Implement mini batch optimization technique to improve the performance of deep learning mode	CO3	PO1,PO2,PO3,PO5,PO9	
8	Demonstrate Convolutional Neural Network with various Convolution functions and Pooling functions	CO4	PO1,PO2,PO3,PO5,PO9	
9	Develop a Residual Network for image classification.	CO5	PO1,PO2,PO3,PO4,PO5,PO9	
10	Build a bidirectional Recurrent Neural Network for any one application	CO5	PO1,PO2,PO3,PO4,PO5,PO9	

Additional Programs				
Deep Learning Lab				
S.No.	Name of the Program	COs	POs	Page No.
1	Implement Gradient Descent With AdaGrad	CO2	PO1,PO2,PO3, PO5,,PO9	
2	Implement Autoencoder using keras	CO3	PO1,PO2,PO3, ,PO5,PO9	

Vision of the Institute

Geethanjali visualizes dissemination of knowledge and skills to students, who eventually contribute to well being of the people of the nation and global community.

Mission of the Institute

- To impart adequate fundamental knowledge in all basic science and engineering technical and Inter-Personals skills so students.
- To bring out creativity in students that would promote innovation, research and entrepreneurship
- To Preserve and promote cultural heritage, humanistic and spiritual values promoting peace and harmony in society

Vision of the Department

To produce globally competent and socially responsible computer science engineers contributing to the advancement of engineering and technology which involves creativity and innovation by providing excellent learning environment with world class facilities.

Mission of the Department

1. To be a center of excellence in instruction, innovation in research and scholarship, and service to the stake holders, the profession, and the public.
2. To prepare graduates to enter a rapidly changing field as a competent computer science engineer.
3. To prepare graduate capable in all phases of software development, possess a firm understanding of hardware technologies, have the strong mathematical background necessary for scientific computing, and be sufficiently well versed in general theory to allow growth within the discipline as it advances.
4. To prepare graduates to assume leadership roles by possessing good communication skills, the ability to work effectively as team members, and an appreciation for their social and ethical responsibility in a global setting.

PROGRAM EDUCATIONAL OBJECTIVES

1. To provide graduates with a good foundation in mathematics, sciences and engineering fundamentals required to solve engineering problems that will facilitate them to find employment in industry and / or to pursue postgraduate studies with an appreciation for lifelong learning.
2. To provide graduates with analytical and problem solving skills to design algorithms, other hardware / software systems, and inculcate professional ethics, inter-personal skills to work in a multi-cultural team.
3. To facilitate graduates to get familiarized with the art software / hardware tools, imbibing creativity and innovation that would enable them to develop cutting-edge technologies of multi-disciplinary nature for societal development.

PROGRAM OUTCOMES (POs)

Program Outcomes (POs) describe what students are expected to know and be able to do by the time of graduation to accomplish Program Educational Objectives (PEOs). The Program Outcomes for Computer Science and Engineering graduates are:

Engineering Graduates would be able to:

PO 1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and The World: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 9: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 10: Project management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 11: Life-Long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO 1: To identify and define the computing requirements appropriate for its solution under given constraints.

PSO 2: To follow the best practices, namely, SEI-CMM levels and 6-sigma which varies from time to time for software development projects using open-ended programming environments to produce software deliverables as per customer needs.

Mapping of Lab Course with Programme Educational Objectives

S.No	Course component	code	Course	Semester	PEO 1	PEO 2	PEO 3
1	Professional core	20CS41L08	Deep Learning Lab	I	√	√	√

Mapping of Lab Course outcomes with Programme outcomes:

CourseCode and Title	PSOs												
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
20CS41L08 – DEEP LEARNING LAB													
CO1. Analyze various learning models.	3	3	3	3	3			3				2	3
CO2. Use feed forward neural networks for learning	3	3	3	3	3			3				2	3
CO3. Highlight the importance of auto encoders and regularization	3	2	2	2	3			2				2	2
CO4. Apply Convolution Neural Networks for learning	3	3	3	3	3			3				2	3
CO5. Apply Recurrent Neural Networks for learning	3	3	3	3	3			3				2	3

Course Objectives and Course Outcomes

DEEP LEARNING LAB

COURSE OBJECTIVES

The main objectives of this course are:

1. Understand various learning models.
2. Learn feed forward neural networks for learning.
3. Learn to use auto encoders and regularization.
4. Understand Convolution Neural Networks for learning.
5. Understand Recurrent Neural Networks for learning.

LAB COURSE OUTCOMES:

At the end of the course, student would be able to

- CO1. Analyze various learning models.
- CO2. Use feed forward neural networks for learning
- CO3. Highlight the importance of auto encoders and regularization
- CO4. Apply Convolution Neural Networks for learning
- CO5. Apply Recurrent Neural Networks for learning

PREREQUISITES:

1. The Student has basic knowledge about basic Machine Learning concepts.
2. The Student has knowledge of writing code in python

INSTRUCTIONS TO THE STUDENTS:

1. Students are required to attend all labs.
2. Students should be dressed in formals when attending the laboratory sessions.
3. Students will work individually in computer laboratories.
4. While coming to the lab bring the observation book and Work book etc.
5. Before coming to the lab, prepare the pre-lab questions. Read through the lab experiment to familiarize you.
6. Utilize 3 hours' time properly to perform the experiment and noting down the outputs.
7. If the experiment is not completed in the prescribed time, the pending work has to be done in the leisure hour or extended hours.
8. You will be expected to submit the completed work book according to the deadlines set up by your instructor.

INSTRUCTIONS TO LABORATORY TEACHERS:

1. Observation book and lab records submitted for the lab work are to be checked and signed before the next lab session.
2. Students should be instructed to switch ON the power supply after the connections are checked by the lab assistant / teacher.
3. The promptness of submission should be strictly insisted by awarding the marks accordingly.
4. Ask viva questions at the end of the experiment.
5. Do not allow students who come late to the lab class.
6. Encourage the students to do the experiments innovatively.
7. Fill continuous Evaluation sheet, on regular basis.
8. Ensure that the students are dressed in formals

SCHEME OF LAB EXAM EVALUATION:**EVALUATION OF INTERNAL MARKS:**

- a) 15 Marks are awarded for day to day work
 - i. Record and Observation book ----- 5Marks
 - ii. Attendance and behavior of student----- 5 Mark
 - iii. Viva and performance ----- 5 Marks
- b) 15 Marks are awarded for conducting laboratory test as follows:
 - 1) Write up and program 5 Marks
 - 2) Execution of Program 5 Marks
 - 3) Viva and performance --- 5 Marks

EVALUATION OF EXTERNAL MARKS:

70 Marks are awarded for conducting laboratory test as follows:

- 1) Algorithm -----25 Marks.
- 2) Write up and program ----- 15 Marks
- 3) Execution of Program -----15 Marks
- 4) Viva.....15 Marks

PERFORMANCE INDICATOR

S.No	Name of Experiment	Date of Exp.	Date of Submission	Marks	Signature	Remarks

PERFORMANCE INDICATOR

S.No	Name of Experiment	Date of Exp.	Date of Submission	Marks	Signature	Remarks

DEEP LEARNING LAB

Date:

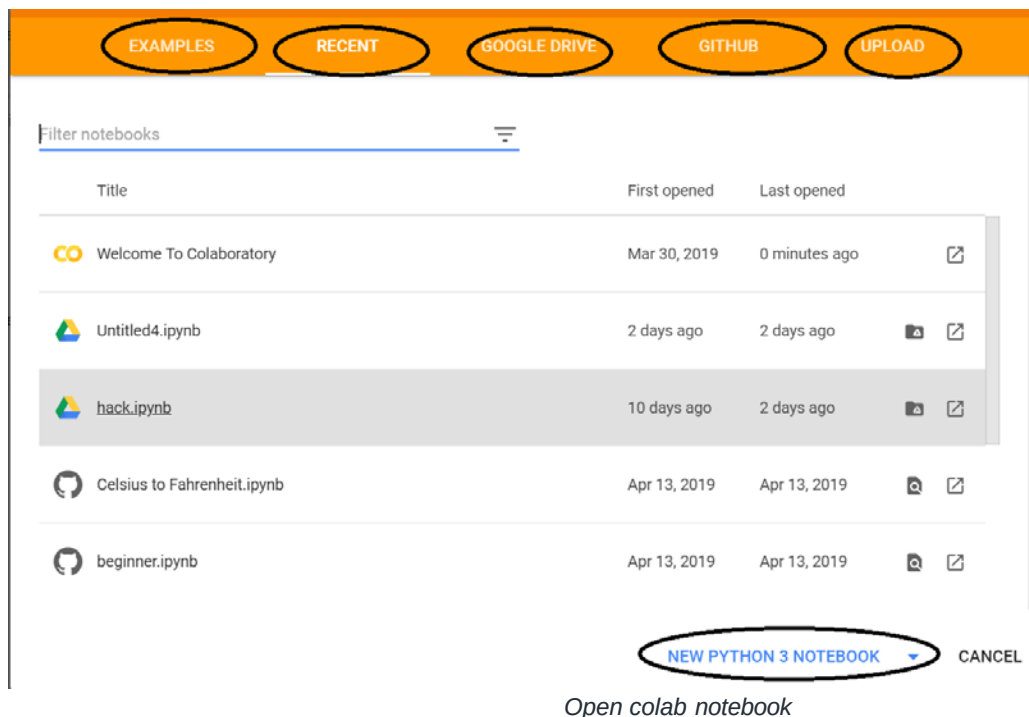
Task 1 : Introduction to IDE

Getting familiar with the usage of Google Colab and using GPU as processing unit

Colaboratory by Google (Google Colab in short) is a Jupyter notebook based runtime environment which allows you to run code entirely on the cloud. This is necessary because it means that you can train large scale ML and DL models even if you don't have access to a powerful machine or a high-speed internet access. Google Colab supports both GPU and TPU instances, which makes it a perfect tool for deep learning and data analytics enthusiasts because of computational limitations on local machines. Since a Colab notebook can be accessed remotely from any machine through a browser, it's well suited for commercial purposes as well. In this tutorial you will learn:

To start working with Colab you first need to log in to your Google account, then go to this link <https://colab.research.google.com>.

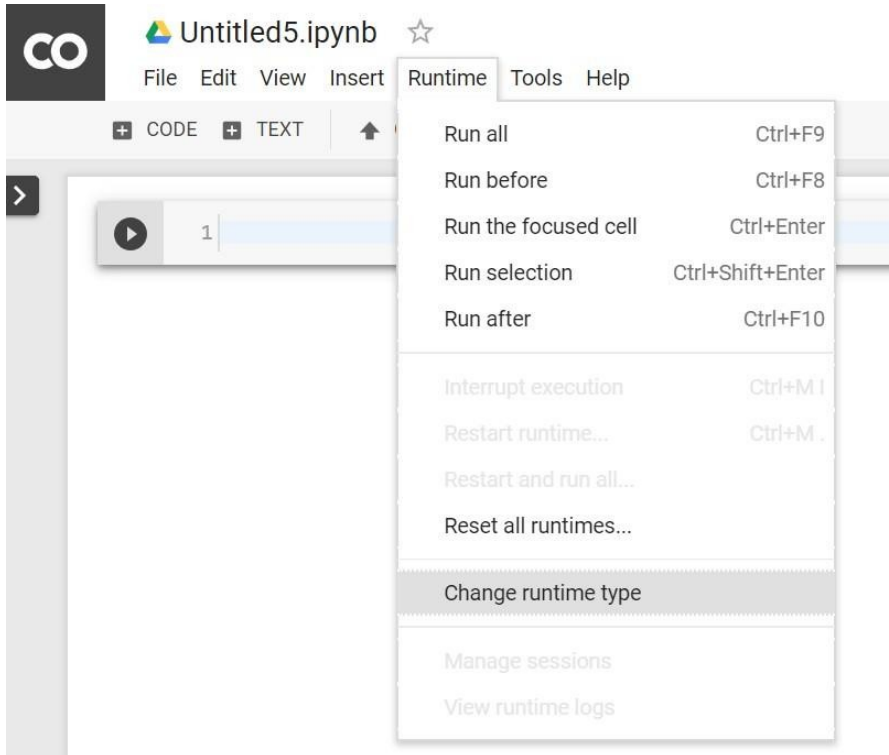
- Opening Jupyter Notebook
- On opening the website you will see a pop-up containing the following tabs –



Date:

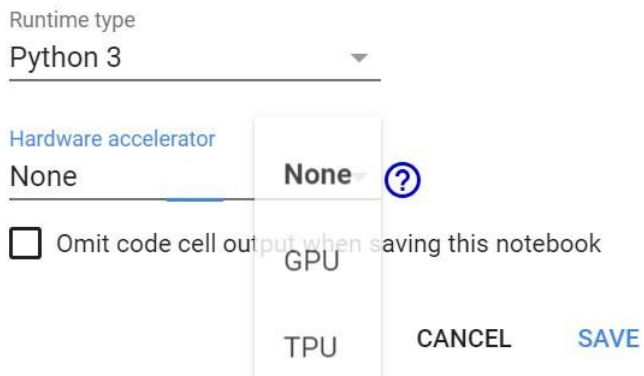
Use GPU and TPU

Click the **“Runtime”** dropdown menu. Select **“Change runtime type”**. Now select anything (GPU, CPU, None) you want in the **“Hardware accelerator”** dropdown menu.



Select **“Change runtime type”**. Select python2 or 3 from the **“Runtime type”** dropdown menu.

Notebook settings



Date:

Week-1

Demonstrate normalization of input data, basic activation functions such as the softmax, sigmoid, dsigmoid, etc.

Workspace:

Output:

Discussion Topics:

- 1. Illustrate the artificial neuron?**
- 2. What is meant by threshold logic unit?**
- 3. What is sigmoid neuron?**
- 4. What is gradient based learning?**

Date:

Week-2

Build a neural network for logistic regression to minimize the cost function and update the parameters

Workspace:

Output:

Discussion Topics:

1. What is logistic regression?

2. What is cost function?

Date:

Week-3

Implement backward propagation neural network for a two- class classification with a single hiddenlayer, non-linear activation function like tanh and compute the cross-entropy loss

Workspace:

Output:

Discussion Topics:

1. What is backpropagation?
2. How to calculate the loss function

Date:

Week-4

Build a deep neural network with more than one hidden layer, non-linear functions like ReLU.

Workspace:

Output:

Discussion Topics:

1. Compare various activation functions

Date:

Week-5

Build deep neural network to any classification problem and compare its accuracy to logistic regression.

Workspace

Output:

Discussion Topics:

1. What is the cost function for logistic regression?

Date:

Week-6

Apply Regularization techniques in deep learning model with backward propagation.

Workspace

Output:

Discussion Topics:

- 1. What is Regularization?**
- 2. What is L1 Regularization?**
- 3. What is L2 regularization?**

Date:

Week-7

Implement mini batch optimization technique to improve the performance of deep learning model.

Workspace:

Output:

Discussion Topics:

- 1. What is Mini-batch Gradient Descent?**
- 2. Compare three gradient descents**
- 3. Write equations to calculate various gradient descents**

Date:

Week-8

Demonstrate Convolutional Neural Network with various Convolution functions and Pooling functions.

Workspace:

Output:

Discussion Topics:

1. What is CNN?
2. What are the layers of CNN
3. Applications of convolutional neural networks

Date:

Week-9

Develop a Residual Network for image classification.

Workspace

Output:

Discussion Topics:

- 1. What is Guided Backpropagation?**
- 2. What are Recurrent Neural Networks?**
- 3. Difference between Recurrent Neural Networks vs. Feedforward Neural Networks?**

Date :

Week-10

Build a bidirectional Recurrent Neural Network for any one application

Workspace

Output

Discussion Topic

1. What is Vanishing and Exploding Gradients?